

# Daily Logic Drill #2

Sheet by David Akinyele-Aje

| Section | Topic                    | Questions | Target time  |
|---------|--------------------------|-----------|--------------|
| A       | Rapid Recognition        | 10        | 2 min 30 sec |
| B       | Manipulation Drills      | 10        | 8 min        |
| C       | Substitution & Structure | 8         | 10 min       |
| D       | Challenge                | 5         | 15 min       |

New toolkit today: Implication, Converse, Inverse & Contrapositive · Necessary vs Sufficient.

Attempt each question before checking answers. Time yourself. No calculators.

## Section A Rapid Recognition

(≤15 s each)

Rapid condition strength: classify as necessary, sufficient, both (equivalent), or neither.

- For real numbers  $x, y$ , classify “ $x > y$ ” as a condition for “ $x^3 > y^3$ ”.
- For non-zero real numbers  $x, y$ , classify “ $x > y$ ” as a condition for “ $\frac{1}{x} < \frac{1}{y}$ ”.
- For the quadratic equation  $x^2 - 2kx + 1 = 0$ , classify “ $k > 1$ ” as a condition for “the equation has two distinct positive real roots”.
- For real numbers  $x, y$ , classify “ $|x + y| = |x| + |y|$ ” as a condition for “ $xy \geq 0$ ”.
- Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be differentiable. Classify “ $f'(x) > 0$  for all  $x \in \mathbb{R}$ ” as a condition for “ $f$  is strictly increasing on  $\mathbb{R}$ ”.
- For real constants  $a, b, c, d$ , classify “ $ad - bc \neq 0$ ” as a condition for “the system  $ax + by = 0$ ,  $cx + dy = 0$  has only the trivial solution  $(x, y) = (0, 0)$ ”.
- For a point  $(x, y) \in \mathbb{R}^2$ , classify “ $x^2 + y^2 \leq 1$ ” as a condition for “ $|x| + |y| \leq \sqrt{2}$ ”.
- For a positive integer  $n$ , classify “ $12 \mid n$ ” as a condition for “ $4 \mid n$  and  $6 \mid n$ ”.
- For a triangle with side lengths  $a, b, c$ , classify “ $a^2 + b^2 = c^2$ ” as a condition for “the area of the triangle is  $\frac{1}{2}ab$ ”.
- For an integer  $n$ , classify “ $n$  is odd” as a condition for “ $n^2 \equiv 1 \pmod{8}$ ”.

## Section B Manipulation Drills

(≤50 s each)

Condition strength on roots, calculus symmetries, and number theory.

- Determine the necessary and sufficient condition on real  $k$  such that the quadratic equation  $x^2 - kx + 9 = 0$  has two distinct real roots that are both strictly greater than 1. (after TMUA 2021 Paper 2 Q9)

2. Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be differentiable. Consider the conditions:
  - I.  $f(-x) = f(x)$  for all  $x \in \mathbb{R}$ .
  - II.  $f'(-x) = -f'(x)$  for all  $x \in \mathbb{R}$ .
 Classify condition I as a condition for condition II. (*after TMUA 2018 Paper 2 Q10*)
3. For positive real numbers  $x, y$ , classify “ $\ln(x + y) < \ln x + \ln y$ ” as a condition for “ $x > 1$  and  $y > 1$ ”.
4. State the contrapositive of: “If  $a^2 + b^2$  is divisible by 3, then both  $a$  and  $b$  are divisible by 3” (for integers  $a, b$ ). Is this contrapositive true?
5. Let  $T_1$  and  $T_2$  be two triangles. Classify “ $T_1$  and  $T_2$  have equal perimeter and equal area” as a condition for “ $T_1$  and  $T_2$  are congruent”.
6. For an integer  $n$ , classify “ $n$  is odd” as a condition for “ $n^3 - n$  is a multiple of 24”.
7. Suppose propositions  $P, Q, R$  satisfy  $P \implies Q$  and  $R \implies Q$ . Must either  $P \implies R$  or  $R \implies P$  be true? Justify with a mathematical counterexample.
8. For real  $c$ , determine whether “ $c < 0$ ” is necessary, sufficient, both, or neither for the quartic equation  $x^4 + x^2 + c = 0$  to have at least two distinct real roots.
9. Consider the statement: “If  $n$  is a multiple of 9, then  $n^2$  is a multiple of 27” (for positive integer  $n$ ). State its converse, and determine whether the converse is true or false.
10. Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be twice differentiable. Classify “ $f'(a) = 0$  and  $f''(a) > 0$ ” as a condition for “ $f$  has a local minimum at  $x = a$ ”.

## Section C Substitution & Structure

( $\leq 90$  s each)

*Chain necessary/sufficient claims, and use the inverse-converse link.*

1. For real numbers  $x, y$ , the condition “ $x + y$  is an integer and  $x - y$  is an integer” is:
  - A) necessary but not sufficient for “ $x$  and  $y$  are both integers”
  - B) sufficient but not necessary for “ $x$  and  $y$  are both integers”
  - C) necessary and sufficient for “ $x$  and  $y$  are both integers”
  - D) neither necessary nor sufficient for “ $x$  and  $y$  are both integers”
2. Determine the condition on real parameters  $p, q$  that is necessary and sufficient for the cubic  $x^3 - 3px + q = 0$  with  $p > 0$  to have three distinct real roots.
  - A)  $p^3 > q^2$
  - B)  $4p^3 > q^2$
  - C)  $p^2 > 4q^3$
  - D)  $q^2 > 4p^3$

3. Classify “a triangle has all three altitudes of equal length” as a condition for “the triangle is equilateral”. Justify using the fact that an altitude to side  $a$  has length  $\frac{2 \times \text{Area}}{a}$ .
4. For positive integers  $a, b, n$ , suppose you’re told only that  $n$  divides  $a$ , or  $n$  divides  $b$  (possibly both). Does this pin down whether  $n$  divides the product  $ab$ ? And working the other way: does knowing  $n$  divides  $ab$  pin down whether  $n$  divides  $a$  or  $n$  divides  $b$ ? (after TMUA 2019 Paper 2 Q5)
- A) “ $n$  divides  $a$  or  $n$  divides  $b$ ” is necessary but not sufficient for “ $n$  divides  $ab$ ”
- B) “ $n$  divides  $a$  or  $n$  divides  $b$ ” is sufficient but not necessary for “ $n$  divides  $ab$ ”
- C) “ $n$  divides  $a$  or  $n$  divides  $b$ ” is necessary and sufficient for “ $n$  divides  $ab$ ”
- D) “ $n$  divides  $a$  or  $n$  divides  $b$ ” is neither necessary nor sufficient for “ $n$  divides  $ab$ ”
5. For a twice-differentiable function  $f : \mathbb{R} \rightarrow \mathbb{R}$ , which of the following is necessary and sufficient for the midpoint convexity condition  $\frac{f(a)+f(b)}{2} \geq f\left(\frac{a+b}{2}\right)$  to hold for all  $a, b \in \mathbb{R}$ ? (after TMUA 2018 Paper 2 Q18)
- A)  $f''(x) > 0$  for all  $x \in \mathbb{R}$
- B)  $f''(x) \geq 0$  for all  $x \in \mathbb{R}$
- C)  $f'(x) \geq 0$  for all  $x \in \mathbb{R}$
- D)  $f(x) \geq 0$  for all  $x \in \mathbb{R}$
6. Classify “ $n^2$  is even” as a condition for “ $n$  is even”.
- A) necessary but not sufficient
- B) sufficient but not necessary
- C) necessary and sufficient
- D) neither necessary nor sufficient
7. Let  $p(x)$  be a cubic polynomial with real coefficients. Which of the following conditions is necessary and sufficient for the existence of  $c \in (a, b)$  such that  $p'(c) = 0$ ? (after TMUA 2021 Paper 2 Q8)
- A)  $p(a) = p(b)$
- B)  $p'(a)p'(b) \leq 0$
- C)  $p'(x) = 0$  has real roots and at least one lies in  $(a, b)$
- D)  $p(a)p(b) < 0$
8. State the contrapositive of “If  $a$  and  $b$  are both irrational, then  $ab$  is irrational.” Is the original statement true? Give a counterexample if false.

## Section D Challenge

(TMUA / SMC difficulty)

*Insight over computation. Each question has a short, elegant solution — find it.*

1. Let  $f(x) = x^2 + bx + c$  with  $f(0) = c$ . Let  $P$  be the statement: “if  $c < 0$ , then  $f$  has two distinct real roots.” Which of the following must be true? (after TMUA 2022 Paper 2 Q5)

- I.  $P$
- II. The converse of  $P$
- III. The contrapositive of  $P$
- A) none of them
- B) I only
- C) II only
- D) III only
- E) I and II only
- F) I and III only
- G) II and III only
- H) I, II and III
2. Consider a finite list of  $n$  real numbers (repeats allowed), sorted into non-decreasing order. Let  $P$  be the statement “ $n$  is odd”, and let  $Q$  be the statement “the middle value of the sorted list equals one of the list’s own entries”.  $P$  is: (after TMUA 2022 Paper 2 Q6)
- A) necessary but not sufficient for  $Q$
- B) sufficient but not necessary for  $Q$
- C) necessary and sufficient for  $Q$
- D) neither necessary nor sufficient for  $Q$
3. For a positive integer  $n$ , let  $P$  be “ $n$  leaves remainder 1 when divided by 4” and  $Q$  be “ $n^2$  leaves remainder 1 when divided by 8.”  $P$  is:
- A) necessary but not sufficient for  $Q$
- B) sufficient but not necessary for  $Q$
- C) necessary and sufficient for  $Q$
- D) neither necessary nor sufficient for  $Q$
4. Statement  $S$ : “If a quadrilateral has four equal sides, then its diagonals are perpendicular.” (a) Is  $S$  true? (b) State  $S$ ’s converse. Is the converse true?
5. Statement  $S$ : “If  $n$  is a positive integer such that  $n^2 - 1$  is prime, then  $n = 2$ .” (a) Is  $S$  true? Prove or disprove. (b) State  $S$ ’s contrapositive. (c) Is “ $n^2 - 1$  is prime” necessary and sufficient for “ $n = 2$ ”?

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*“Speed comes from recognition, not from rushing. Every shortcut is a pattern you have internalised.”*

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Found a mistake or want to contribute a sheet?  
Visit the open-source project at [github.com/The-CerealDev/speed-maths](https://github.com/The-CerealDev/speed-maths)